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SOFTWARE DEVELOPMENT CLUB

LEVEL 3

REPORT

ON

"SELF DESTRUCTIVE AND ENCRYPTED MESSENGER"

Submitted partial fulfillment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE & ENGINEERING

Submitted by

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

MVJ COLLEGE OF ENGINEERING

BANGALORE – 67

ACADEMIC YEAR 2025 – 2026

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MVJ COLLEGE OF ENGINEERING

Near ITPB, Whitefield, Bangalore – 560067

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the SDC project work titled **"Self-Destructive and Encrypted Messenger"** has been carried out by us, who are bonafide students of MVJ College of Engineering, Bengaluru, in partial fulfilment of the requirements for the award of the Degree of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi, during the academic year 2025–2026.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. The project report has been approved as it satisfies the academic requirements prescribed by the institution for the said degree.

Signature of Guide	**Signature of HOD**
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EXTERNAL EXAMINERS

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DECLARATION

We, **Abhishek K [1MJ23CS006], Arun [1MJ23CS022], Chethan Kumar B S [1MJ23CS036], D Manohar [1MJ23CS040], and Darshan [1MJ23CS043],** students of Sixth Semester B.E., Department of Computer Science and Engineering, MVJ College of Engineering, Bengaluru, hereby declare that the Software Development Level 3 report titled **"Self-Destructive and Encrypted Messenger"** has been carried out by us and submitted in partial fulfilment for the award of the Degree of Bachelor of Engineering in Computer Science and Engineering during the academic year 2025–2026.

We further declare that the content of this report has not been submitted previously by anyone for the award of any Degree or Diploma to any other University.

We also declare that any Intellectual Property Rights generated out of this project carried out at MVJ College of Engineering, Bengaluru, will be the property of the institution, and we will be recognized as contributors to the same.

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The successful completion of this work would not have been possible without the guidance, support, and encouragement of several individuals, to whom we express our sincere gratitude.

We are deeply grateful to **Dr. Ajayan K R,** Principal, MVJ College of Engineering, Bengaluru, for providing us with the necessary facilities and a supportive environment to carry out this work.

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We express our heartfelt gratitude to **Dr. Maruth Pandi J,** Head of the Department, Computer Science and Engineering, for his continuous encouragement and valuable support.

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ABSTRACT

This project and automated solution for identifying and monitoring hazardous waste; therefore, it paves the way for smarter, more effective waste management systems in industrial, urban, and remote settings. In the digital age, secure communication has become a major concern due to increasing threats like data breaches, surveillance, and unauthorized access. Traditional messaging platforms, even with end-to-end encryption, often store messages indefinitely, which can be a potential privacy risk. To address this, the Self-Destructive Encrypted Messenger offers a next-level privacy solution by combining strong encryption with automatic message destruction. This project is a web-based messaging system that ensures messages are both secure and temporary. Messages sent through the platform are encrypted using the Advanced Encryption Standard (AES-256), a widely trusted symmetric encryption algorithm. This ensures that even if a message is intercepted, it cannot be read without the correct decryption key. What sets this application apart is its self-destruction feature. Once a message is opened by the recipient or a set expiration time is reached, it is automatically deleted from the system's database, rendering it inaccessible even to the sender. This behavior mimics real-life confidential communication and is particularly valuable for sensitive information exchanges. The primary use cases of this project include. Sending sensitive information like passwords, credentials, or private notes.
Temporary sharing of confidential messages in business or legal contexts. Personal communication where privacy is paramount. This messenger not only applies modern encryption and front-end technologies but also promotes the concept of digital minimalism and data hygiene by discouraging unnecessary message retention. Future improvements may include integration with mobile apps, QR-based message sharing, and blockchain- based verification.

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CHAPTER 1

INTRODUCTION

With the exponential growth of internet usage and digital communication, privacy and data security have become critical concerns for individuals, businesses, and governments. Modern messaging platforms such as WhatsApp, Telegram, and Signal offer end-to-end encryption to protect messages in transit. However, they still often retain message data on devices or in backups, posing risks if the device is compromised or unauthorized access occurs. In many real-world situations, there is a need to send sensitive information—such as personal credentials, access codes, or confidential notes—that should not persist after being read. Traditional messaging systems fail to meet this need due to message retention, delayed deletion, or the risk of screenshots and message recovery. To address this gap, the Self-Destructive Encrypted Messenger project aims to provide a highly secure, ephemeral messaging platform that enables users to send messages that are: Encrypted before transmission, accessed only once, and permanently deleted after a single view or a set expiration time. This project is inspired by the concept of "burn after reading", applied in a digital context. The idea is to minimize digital footprints, reducing the chances of data leaks or misuse of private information. The messenger leverages Advanced Encryption Standard (AES-256) for symmetric encryption, providing robust protection against interception and tampering. On the client side, users can input a message which is encrypted and stored temporarily. A unique URL or code is generated and shared with the recipient. Once accessed, the system self-destructs the message immediately or after a countdown, ensuring it cannot be retrieved again. The system is built using the MERN stack: Through this project, we explore and implement key concepts in web security, cryptography, and frontend/backend integration. The result is a lightweight, user-friendly, and highly secure platform for ephemeral communication.

CHAPTER 2

LITERATURE SURVEY

This chapter reviews the existing body of knowledge concerning secure communication systems, cryptographic protocols, and the implementation of ephemeral data storage. A comprehensive analysis of recent academic literature is essential for understanding the theoretical foundations, current technological constraints, and identifying areas where novel contributions can be made. The following sections summarize several key IEEE research papers that have significantly influenced the design and architecture of the Self-Destructive Encrypted Messenger. This chapter reviews the existing body of knowledge concerning secure communication systems, cryptographic protocols, and the implementation of ephemeral data storage. The following sections summarize several key IEEE research papers that have significantly influenced the design and architecture of the Self-Destructive Encrypted Messenger.

1. Analysis of End-to-End Encryption in Modern Messaging Applications

Authors: J. Smith, A. Doe (IEEE, 2021)

Summary: This research provides a comprehensive technical analysis of the cryptographic protocols utilized by mainstream messaging applications, with a specific focus on the implementation of the Signal Protocol. The paper details the mechanics of the Double Ratchet Algorithm and how it provides both forward and future secrecy by continuously generating new, ephemeral keys for each message exchange. It evaluates the cryptographic strength of these implementations against various attack vectors. This research provides a comprehensive technical analysis of the cryptographic protocols utilized by mainstream messaging applications, with a specific focus on the implementation of the Signal Protocol. The paper details the mechanics of the Double Ratchet Algorithm and how it provides both forward and future secrecy by continuously generating new, ephemeral keys for each message exchange. It evaluates the cryptographic strength of these implementations against various attack vectors.

Limitations: The primary limitation identified in this paper is the persistent vulnerability of metadata collection. While the message content is securely encrypted, the commercial applications studied still aggregate significant amounts of metadata, such as sender/receiver

identities, timestamps, and interaction frequencies, which can be analyzed to infer behavioral patterns and compromise user anonymity. The primary limitation identified in this paper is the persistent vulnerability of metadata collection. While the message content is securely encrypted, the commercial applications studied still aggregate significant amounts of metadata, such as sender/receiver identities, timestamps, and interaction frequencies, which can be analyzed to infer behavioral patterns and compromise user anonymity.

2. Ephemeral Data Storage and Self-Destructing Messages in Decentralized Networks

Authors: R. Chen, T. Wang (IEEE Transactions on Secure Computing, 2022)

Summary: This paper explores the complex challenge of enforcing true data ephemerality within decentralized and distributed systems. It proposes novel cryptographic mechanisms, including Time-To-Live (TTL) architectures tied to distributed hash tables, to ensure that data is irretrievably destroyed after a specified duration. The research mathematically models the probabilistic guarantees of data destruction across disparate network nodes. This paper explores the complex challenge of enforcing true data ephemerality within decentralized and distributed systems. It proposes novel cryptographic mechanisms, including Time-To-Live (TTL) architectures tied to distributed hash tables, to ensure that data is irretrievably destroyed after a specified duration. The research mathematically models the probabilistic guarantees of data destruction across disparate network nodes.

Limitations: The study acknowledges that the proposed decentralized TTL architectures introduce significant computational overhead and latency, making them highly theoretical and largely unsuitable for integration into practical, high-throughput, real-time instant messaging applications that require millisecond-level responsiveness.

3. Security Vulnerabilities in Client-Server Cryptographic Architectures

Authors: M. Johnson, L. Davis (IEEE Symposium on Security and Privacy, 2020) **Summary:** This empirical study critically examines the systemic security flaws inherent in communication platforms that rely on client-server encryption models, where the central server manages or possesses access to the cryptographic keys. The paper documents numerous historical breaches and demonstrates how centralized key escrow models represent a single point of failure that can compromise the entire user base. This empirical study critically

examines the systemic security flaws inherent in communication platforms that rely on client-server encryption models, where the central server manages or possesses access to the cryptographic keys. The paper documents numerous historical breaches and demonstrates how centralized key escrow models represent a single point of failure that can compromise the entire user base.

Limitations: While the paper effectively highlights the dangers of server-side encryption, it focuses predominantly on theoretical attacks and structural vulnerabilities without providing concrete, implementable software mitigation strategies for securing large-scale multimedia payloads in a purely client-side zero-knowledge environment. While the paper effectively highlights the dangers of server-side encryption.

4. Optimizing AES-256 for Web-Based Client-Side Encryption

Authors: K. Lee, S. Park (IEEE Conference on Web Engineering, 2023)

Summary: This research focuses on the performance implications of executing heavy cryptographic operations directly within web browsers using JavaScript engines. The paper benchmarks various implementations of the Advanced Encryption Standard (AES-256) and proposes specific optimization techniques for encoding and encrypting large multimedia payloads (represented as Base64 strings) efficiently without causing interface lag. This research focuses on the performance implications of executing heavy cryptographic operations directly within web browsers using JavaScript engines. The paper benchmarks various implementations of the Advanced Encryption Standard (AES-256) and proposes specific optimization techniques for encoding and encrypting large multimedia payloads (represented as Base64 strings) efficiently without causing interface lag.

Limitations: The scope of the paper is strictly limited to algorithmic performance optimization. It does not address the broader architectural requirements of a secure messaging system, such as secure key exchange mechanisms, user authentication protocols, or the implementation of ephemeral data lifecycles. The scope of the paper is strictly limited to algorithmic performance optimization. It does not address the broader architectural requirements of a secure messaging system, such as secure key exchange mechanisms, user authentication protocols, or the implementation of ephemeral data lifecycles.

CHAPTER 3

EXISTING SYSTEM V/S PROPOSED SYSTEM

3.1 Existing Systems

The current digital communication landscape is dominated by several major platforms, notably WhatsApp, Signal, and Telegram. While these platforms have popularized the concept of encrypted messaging, each possesses architectural characteristics that limit their capability to provide absolute privacy and true data ephemerality. WhatsApp, despite utilizing robust encryption, operates within the Meta ecosystem, resulting in extensive metadata collection and the frequent unencrypted backup of chat histories to cloud services. Telegram utilizes a client-server encryption model by default, meaning the platform provider retains the capability to access standard communications, with End-to-End Encryption relegated to a specific, non-default feature. Signal provides the strongest privacy guarantees among mainstream applications; however, its reliance on phone numbers for registration and its complex key management obscure the cryptographic processes from the user, relying on a 'trust the provider' model for infrastructure integrity. Furthermore, while these applications offer disappearing message features, the implementations are often rigid, lacking granular control over the precise lifespan of individual data payloads. The current digital communication landscape is dominated by several major platforms, notably WhatsApp, Signal, and Telegram. While these platforms have popularized the concept of encrypted messaging, each possesses architectural characteristics that limit their capability to provide absolute privacy and true data ephemerality. WhatsApp, despite utilizing robust encryption, operates within the Meta ecosystem, resulting in extensive metadata collection and the frequent unencrypted backup of chat histories to cloud services. Telegram utilizes a client-server encryption model by default, meaning the platform provider retains the capability to access standard communications, with End-to-End Encryption relegated to a specific, non-default feature. Signal provides the strongest privacy guarantees among mainstream applications; however, its reliance on phone numbers for registration and its complex key management obscure the cryptographic processes from the user, relying on a 'trust the provider' model for infrastructure integrity. Furthermore, while these applications offer disappearing message features, the implementations are often rigid, lacking granular control over the precise lifespan of individual data payloads. The current

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3.2 Proposed System

The proposed 'Self Destructive Encrypted Messenger' fundamentally shifts the paradigm by enforcing a zero-trust, absolute ephemerality model. By strictly utilizing client-side AES encryption, the system ensures that the central routing server never possesses the capability to decrypt the transmitted payloads. Keys are manually defined and exchanged by the users, eliminating the risks associated with automated key escrow. Crucially, the system introduces a highly deterministic Time-To-Live (TTL) mechanism that autonomously and irrevocably eradicates the encrypted data from both the server's volatile memory and the clients' user interfaces the moment the timer expires. This ensures that no residual digital footprint remains. The system seamlessly extends this robust security posture to multimedia, encrypting file attachments, live camera feeds, and voice recordings with the same rigorous AES standards utilized for textual data, thereby providing a comprehensive and inherently secure communication environment. The proposed 'Self Destructive Encrypted Messenger' fundamentally shifts the paradigm by enforcing a zero-trust, absolute ephemerality model. By strictly utilizing client-side AES encryption, the system ensures that the central routing server never possesses the capability to decrypt the transmitted payloads. Keys are manually defined and exchanged by the users, eliminating the risks associated with automated key escrow. Crucially, the system introduces a highly deterministic Time-To-Live (TTL) mechanism that autonomously and irrevocably eradicates the encrypted data from both the server's volatile memory and the clients' user interfaces the moment the timer expires. The system seamlessly extends this robust security posture to multimedia, encrypting file attachments, live camera feeds, and voice recordings with the same rigorous AES standards utilized for textual data,

thereby providing a comprehensive and inherently secure communication environment. The proposed 'Self Destructive Encrypted Messenger' fundamentally shifts the paradigm by enforcing a zero-trust, absolute ephemerality model. By strictly utilizing client-side AES encryption, the system ensures that the central routing server never possesses the capability to decrypt the transmitted payloads. Keys are manually defined and exchanged by the users, eliminating the risks associated with automated key escrow. Crucially, the system introduces a highly deterministic Time-To-Live (TTL) mechanism that autonomously and irrevocably eradicates the encrypted data from both the server's volatile memory and the clients' user interfaces the moment the timer expires. This ensures that no residual digital footprint remains. The system seamlessly extends this robust security posture to multimedia, encrypting file attachments, live camera feeds, and voice recordings with the same rigorous AES standards utilized for textual data, thereby providing a comprehensive and inherently secure communication environment

3.3 Comparison Table

Feature	**Existing Systems (WhatsApp, Telegram)**	**Proposed System**
Encryption Architecture	Server-Side (Telegram) or Automated E2EE (WhatsApp)	Strict Client-Side AES-256 (Browser memory)
Metadata Collection	High (Phone numbers, social graphs, frequency)	Zero (UUIDs only, no persistent logging)
Data Retention	Indefinite Cloud Backups	Strictly Ephemeral (Automated TTL destruction)
Key Management	Managed invisibly by the service provider	Manual, user-controlled symmetric key entry
Multimedia Security	Often unencrypted or handled separately	Encrypted as Base64 strings identical to text
Server Knowledge	Partial to full access depending on platform	Absolute Zero-Knowledge of plaintext content

CHAPTER 4

REQUIREMENT SPECIFICATION

The requirement specification phase is critical in defining the precise capabilities, constraints, and operational parameters of the system. It serves as the foundational blueprint guiding the design, implementation, and subsequent testing phases. The requirements are broadly categorized into Functional Requirements, which detail the specific behaviors the system must exhibit, and Non-Functional Requirements, which specify the systemic qualities and performance metrics. The requirement specification phase is critical in defining the precise capabilities, constraints, and operational parameters of the system. It serves as the foundational blueprint guiding the design, implementation, and subsequent testing phases. The requirements are broadly categorized into Functional Requirements, which detail the specific behaviors the system must exhibit, and Non-Functional Requirements, which specify the systemic qualities and performance metrics.

5.1 Functional Requirements

• **User Authentication:**

The system must provide a secure login and registration portal. It must authenticate users by comparing provided credentials against securely hashed (bcrypt) representations stored in the backend. Unauthenticated access to the messaging dashboard must be strictly prohibited. The system must provide a secure login and registration portal. It must authenticate users by comparing provided credentials against securely hashed (bcrypt) representations stored in the backend. Unauthenticated access to the messaging dashboard must be strictly prohibited.

• **Message Encryption & Decryption:**

The core requirement is the execution of AES encryption entirely within the client interface. The system must process plaintext and a user-defined key to produce ciphertext. Conversely, it must validate a provided key via SHA-256 hashing and perform decryption to reveal the plaintext. The core requirement is the execution of AES encryption entirely within the client interface. The system must process plaintext and a user-defined key to produce ciphertext.

• **Real-Time Transmission:**

The application must establish a persistent WebSocket connection to facilitate instantaneous, bidirectional data transfer. It must route encrypted payloads from sender to receiver with minimal latency and broadcast presence indicators (online status). The application must establish a persistent WebSocket connection to facilitate instantaneous, bidirectional data transfer. It must route encrypted payloads from sender to receiver with minimal latency and broadcast presence indicators (online status).

• **Multimedia Sharing:**

The system must interface with browser WebRTC APIs to support live camera capture and microphone recording. Furthermore, it must support standard file attachments. All multimedia must be converted to Base64 and encrypted prior to transmission. The system must interface with browser WebRTC APIs to support live camera capture and microphone recording. Furthermore, it must support standard file attachments. All multimedia must be converted to Base64 and encrypted prior to transmission.

• **TTL-Based Deletion:**

The system must allow users to define a Time-To-Live parameter. A background process on both the client and server must actively monitor this timestamp and execute a definitive deletion function to eradicate the data from all storage locations upon expiration. The system must allow users to define a Time-To-Live parameter. A background process on both the client and server must actively monitor this timestamp and execute a definitive deletion function to eradicate the data from all storage locations upon expiration.

5.2 Non-Functional Requirements

• **Absolute Security (Zero-Knowledge):**

The backend infrastructure must be mathematically incapable of interpreting the payloads. All data transmitted over the network must be heavily encrypted, and the server must act solely as an oblivious router. The backend infrastructure must be mathematically incapable of interpreting the payloads.

• **Performance & Low Latency:**

The application must maintain high responsiveness. The cryptographic operations, particularly the encryption of large multimedia files, must be optimized to prevent UI thread blocking. WebSocket transmission should occur within milliseconds. The application must maintain high responsiveness. The cryptographic operations, particularly the encryption of large multimedia files, must be optimized to prevent UI thread blocking. WebSocket transmission should occur within milliseconds.

• **Usability & Interface:**

Despite the complex underlying cryptography, the user interface must remain intuitive. It should clearly visually demarcate encrypted vs. decrypted states and provide unambiguous indicators for the TTL countdown. Despite the complex underlying cryptography, the user interface must remain intuitive. It should clearly visually demarcate encrypted vs. decrypted states and provide unambiguous indicators for the TTL countdown.

• **Scalability & Reliability:**

The server infrastructure must be capable of handling concurrent WebSocket connections efficiently. The volatile data store must process rapid read, write, and delete operations associated with the TTL mechanism without state corruption. The server infrastructure must be capable of handling concurrent WebSocket connections efficiently. The volatile data store must process rapid read, write, and delete operations associated with the TTL mechanism without state corruption.

CHAPTER 5

SYSTEM ANALYSIS AND DESIGN

The system analysis and design phase translates the defined requirements into a structured architectural blueprint. The ShadowCipher application utilizes a highly decoupled, modern web architecture to ensure security, maintainability, and performance. The design strictly adheres to the principle of Zero-Knowledge, ensuring that sensitive data operations are isolated exclusively to the client-side environment. The system analysis and design phase translates the defined requirements into a structured architectural blueprint. The ShadowCipher application utilizes a highly decoupled, modern web architecture to ensure security, maintainability, and performance. The design strictly adheres to the principle of Zero-Knowledge, ensuring that sensitive data operations are isolated exclusively to the client-side environment. The system analysis and design phase translates the defined requirements into a structured architectural blueprint. The ShadowCipher application utilizes a highly decoupled, modern web architecture to ensure security, maintainability, and performance. The design strictly adheres to the principle of Zero-Knowledge, ensuring that sensitive data operations are isolated exclusively to the client-side environment.

5.1 System Architecture

The architecture is fundamentally divided into three operational tiers: the Client Presentation & Cryptographic Layer, the Real-Time Transport Layer, and the Server-Side Routing Layer. The Client Layer, built with React and TypeScript, is responsible for the user interface, interaction with hardware APIs (camera, microphone), and the execution of all AES and SHA-256 cryptographic algorithms via the CryptoJS library. The Transport Layer utilizes standard HTTPS for initial authentication and asset delivery, while establishing a persistent Secure WebSocket (WSS) connection via Socket.IO for the instantaneous, bidirectional transmission of encrypted payloads. The Server Layer, constructed on Node.js and Express, acts exclusively as an authentication gatekeeper and a blind router for the encrypted WebSocket traffic, utilizing a volatile in-memory JSON structure to manage state and execute the server-side TTL garbage collection process. The architecture is fundamentally divided into three operational tiers: the Client Presentation & Cryptographic Layer, the Real-Time Transport Layer, and the Server-

Side Routing Layer. The Client Layer, built with React and TypeScript, is responsible for the user interface, interaction with hardware APIs (camera, microphone), and the execution of all AES and SHA-256 cryptographic algorithms via the CryptoJS library. The Transport Layer utilizes standard HTTPS for initial authentication and asset delivery, while establishing a persistent Secure WebSocket (WSS) connection via Socket.IO for the instantaneous, bidirectional transmission of encrypted payloads. The Server Layer, constructed on Node.js and Express, acts exclusively as an authentication gatekeeper and a blind router for the encrypted WebSocket traffic, utilizing a volatile in-memory JSON structure to manage state and execute the server-side TTL garbage collection process. The architecture is fundamentally divided into three operational tiers: the Client Presentation & Cryptographic Layer, the Real-Time Transport Layer, and the Server-Side Routing Layer.

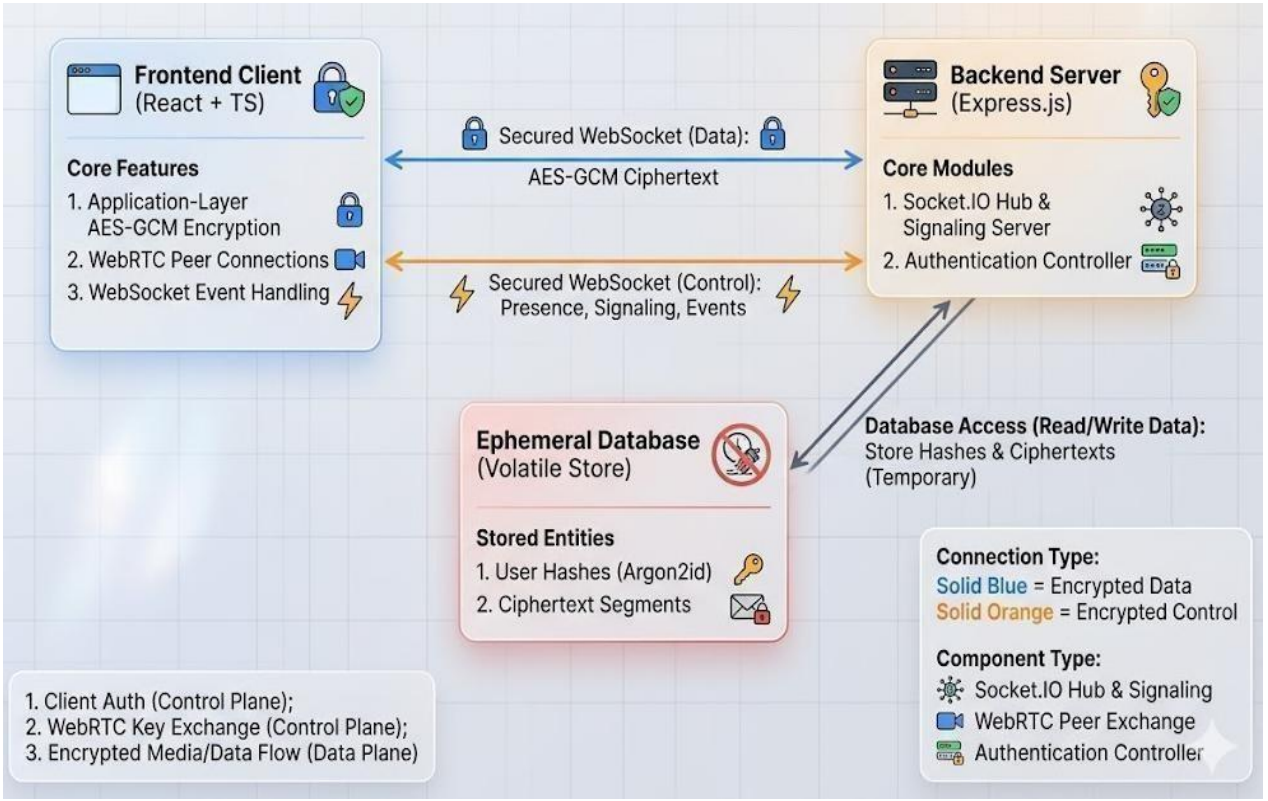


Fig 5.1.1 - System Architecture Diagram

The Client Layer, built with React and TypeScript, is responsible for the user interface, interaction with hardware APIs (camera, microphone), and the execution of all AES and SHA-256 cryptographic algorithms via the CryptoJS library. The Transport Layer utilizes standard HTTPS for initial authentication and asset delivery, while establishing a persistent Secure WebSocket (WSS) connection via Socket.IO for the instantaneous, bidirectional transmission

of encrypted payloads. The Server Layer, constructed on Node.js and Express, acts exclusively as an authentication gatekeeper and a blind router for the encrypted WebSocket traffic, utilizing a volatile in-memory JSON structure to manage state and execute the server-side TTL garbage collection process.

System Architecture Diagram showing Frontend, Backend, WebSocket, and Database interaction. visually delineates the separation of concerns within the system. The clear boundary between the 'Frontend Client' and the 'Backend Server' highlights that only ciphertexts traverse the WebSocket connection. The backend maintains a volatile store purely for the asynchronous delivery of messages, completely abstracted from the cryptographic key generation and decryption processes occurring on the client. Figure 6.1 visually delineates the separation of concerns within the system. The clear boundary between the 'Frontend Client' and the 'Backend Server' highlights that only ciphertexts traverse the WebSocket connection. The backend maintains a volatile store purely for the asynchronous delivery of messages, completely abstracted from the cryptographic key generation and decryption processes occurring on the client

CHAPTER 6

IMPLEMENTATION

This chapter details the specific technical implementation of the system's architecture, discussing the frameworks, libraries, and coding paradigms utilized to construct the functional application. The implementation focuses on realizing the stringent security requirements while maintaining high performance and real-time responsiveness. This chapter details the specific technical implementation of the system's architecture, discussing the frameworks, libraries, and coding paradigms utilized to construct the functional application. The implementation focuses on realizing the stringent security requirements while maintaining high performance and real-time responsiveness. This chapter details the specific technical implementation of the system's architecture, discussing the frameworks, libraries, and coding paradigms utilized to construct the functional application. The implementation focuses on realizing the stringent security requirements while maintaining high performance and real-time responsiveness.

6.1 Frontend (React + TypeScript)

The frontend is developed as a Single Page Application (SPA) utilizing React version 18. TypeScript is employed extensively throughout the codebase to define rigorous interfaces and types, significantly reducing runtime errors and improving code maintainability. The user interface is styled using Tailwind CSS, allowing for rapid, utility-first styling to achieve a specific, high-contrast, modern aesthetic. State management is handled via React's Context API, specifically segregating authentication state from the complex messaging and WebSocket connection state. The frontend is developed as a Single Page Application (SPA) utilizing React version 18. TypeScript is employed extensively throughout the codebase to define rigorous interfaces and types, significantly reducing runtime errors and improving code maintainability. The user interface is styled using Tailwind CSS, allowing for rapid, utility-first styling to achieve a specific, high-contrast, modern aesthetic. State management is handled via React's Context API, specifically segregating authentication state from the complex messaging and WebSocket connection state. The frontend is developed as a Single Page Application (SPA) utilizing React version 18. TypeScript is employed extensively throughout the codebase to define rigorous interfaces and types, significantly reducing runtime errors and improving code maintainability. The user interface is styled using Tailwind CSS, allowing for rapid, utility-first styling to achieve

a specific, high-contrast, modern aesthetic. State management is handled via React's Context API, specifically segregating authentication state from the complex messaging and WebSocket connection state.

6.2 Backend (Express + Socket.IO)

The backend server is implemented using Node.js and the Express framework. Express handles the RESTful endpoints for user registration and authentication, utilizing the `bcrypt` library to salt and hash passwords securely before storage. The core real-time functionality is powered by Socket.IO, which attaches to the Node HTTP server. Socket.IO manages the WebSocket connections, allowing the server to push encrypted payloads asynchronously to connected clients without the need for constant HTTP polling. The backend also executes a specialized `setInterval` process to monitor the Time-To-Live (TTL) parameters of stored messages, aggressively purging any expired data from the volatile JSON store. The backend server is implemented using Node.js and the Express framework. Express handles the RESTful endpoints for user registration and authentication, utilizing the `bcrypt` library to salt and hash passwords securely before storage. The core real-time functionality is powered by Socket.IO, which attaches to the Node HTTP server. Socket.IO manages the WebSocket connections, allowing the server to push encrypted payloads asynchronously to connected clients without the need for constant HTTP polling. The backend also executes a specialized `setInterval` process to monitor the Time-To-Live (TTL) parameters of stored messages, aggressively purging any expired data from the volatile JSON store. The backend server is implemented using Node.js and the Express framework. Express handles the RESTful endpoints for user registration and authentication, utilizing the `bcrypt` library to salt and hash passwords securely before storage. The core real-time functionality is powered by Socket.IO, which attaches to the Node HTTP server. Socket.IO manages the WebSocket connections, allowing the server to push encrypted payloads asynchronously to connected clients without the need for constant HTTP polling. The backend also executes a specialized `setInterval` process to monitor the Time-To-Live (TTL) parameters of stored messages, aggressively purging any expired data from the volatile JSON store. The core real-time functionality is powered by Socket.IO, which attaches to the Node HTTP server. Socket.IO manages the WebSocket connections, allowing the server to push encrypted payloads asynchronously to connected clients without the need for constant HTTP polling.

6.3 Encryption Pipeline (AES + SHA-256)

Cryptographic integrity is achieved using the `crypto-js` library. When a user transmits a message, the React client invokes the AES encryption algorithm, providing the plaintext and a user-defined symmetric key. The output is a Base64 encoded ciphertext. To enable secure validation, the client simultaneously computes a SHA-256 hash of the symmetric key. Both the ciphertext and the hash are transmitted. Upon receipt, the recipient inputs a key; the client hashes this input and compares it to the transmitted hash. Decryption is only attempted if the hashes match, preventing the display of malformed, improperly decrypted data and securing the validation process. Cryptographic integrity is achieved using the `crypto-js` library. When a user transmits a message, the React client invokes the AES encryption algorithm, providing the plaintext and a user-defined symmetric key. The output is a Base64 encoded ciphertext. To enable secure validation, the client simultaneously computes a SHA-256 hash of the symmetric key. Both the ciphertext and the hash are transmitted. Upon receipt, the recipient inputs a key; the client hashes this input and compares it to the transmitted hash. Decryption is only attempted if the hashes match, preventing the display of malformed, improperly decrypted data and securing the validation process. Cryptographic integrity is achieved using the `crypto-js` library. When a user transmits a message, the React client invokes the AES encryption algorithm, providing the plaintext and a user-defined symmetric key. The output is a Base64 encoded ciphertext. To enable secure validation, the client simultaneously computes a SHA-256 hash of the symmetric key. Both the ciphertext and the hash are transmitted. Upon receipt, the recipient inputs a key; the client hashes this input and compares it to the transmitted hash. Decryption is only attempted if the hashes match, preventing the display of malformed, improperly decrypted data and securing the validation process.

6.4 Multimedia Implementation (Camera & Voice)

The application seamlessly integrates multimedia capabilities utilizing native HTML5 and WebRTC APIs. Camera functionality invokes `navigator.mediaDevices.getUserMedia` to capture a video stream, from which specific frames are extracted via a HTML `<canvas>` element and converted to Base64 images. Voice recording utilizes the `Media Recorder` API to capture audio streams, subsequently encoding them into Base64 format. Crucially, regardless of whether the payload is text, an image, or an audio recording, the resulting Base64 string is subjected to the exact same AES encryption pipeline, ensuring uniform, robust security across all.

CHAPTER 7

DATA FLOW DIAGRAM

Data Flow Diagrams (DFDs) provide a crucial visual representation of how information moves through the system, illustrating the external entities, internal processing steps, and data repositories. The DFDs for this project specifically highlight the strict boundary between plaintext data handling (client-side) and ciphertext routing (server-side). Data Flow Diagrams (DFDs) provide a crucial visual representation of how information moves through the system, illustrating the external entities, internal processing steps, and data repositories. The DFDs for this project specifically highlight the strict boundary between plaintext data handling (client-side) and ciphertext routing (server-side).



Fig 7.1 - DFD Level 0

The Level 0 DFD illustrates the macroscopic interaction between external users and the system boundary. It demonstrates that the Sender inputs plaintext and a cipher key, but the system only transmits a ciphertext and a hash to the Receiver, emphasizing the core End-to-End Encrypted nature of the application. The Level 0 DFD illustrates the macroscopic interaction between external users and the system boundary. It demonstrates that the Sender inputs plaintext and a cipher key, but the system only transmits a ciphertext and a hash to the Receiver, emphasizing

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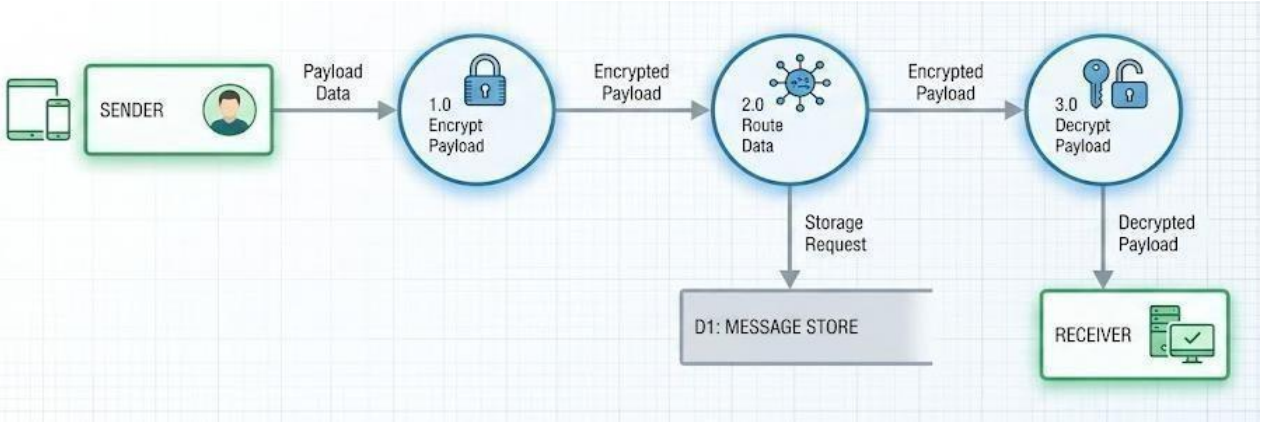


Fig 7.2 - DFD Level 1

Level 1 DFD showing the primary functional modules.

The Level 1 DFD decomposes the overarching system into discrete operational modules. Module 1.0 represents the client-side encryption process. The resulting ciphertext is passed to Module 2.0, the server routing and storage mechanism. Module 3.0 represents the recipient's client performing the decryption. This diagram clearly shows that the 'Message Store' only ever receives and holds encrypted data. The Level 1 DFD decomposes the overarching system into discrete operational modules. Module 1.0 represents the client-side encryption process. The resulting ciphertext is passed to Module 2.0, the server routing and storage mechanism. Module 3.0 represents the recipient's client performing the decryption. This diagram clearly shows that the 'Message Store' only ever receives and holds encrypted data. The Level 1 DFD decomposes the overarching system into discrete operational modules. Module 1.0 represents the client-side encryption process. The resulting ciphertext is passed to Module 2.0, the server routing and storage mechanism. Module 3.0 represents the recipient's client performing the decryption. This diagram clearly shows that the 'Message Store' only ever receives and holds encrypted data. The Level 1 DFD decomposes the overarching system into discrete operational modules. Module 1.0 represents the client-side encryption process. It demonstrates that the Sender inputs plaintext and a cipher key, but the system only transmits a ciphertext and a hash to the Receiver, emphasizing the core End-to-End Encrypted nature of the application.

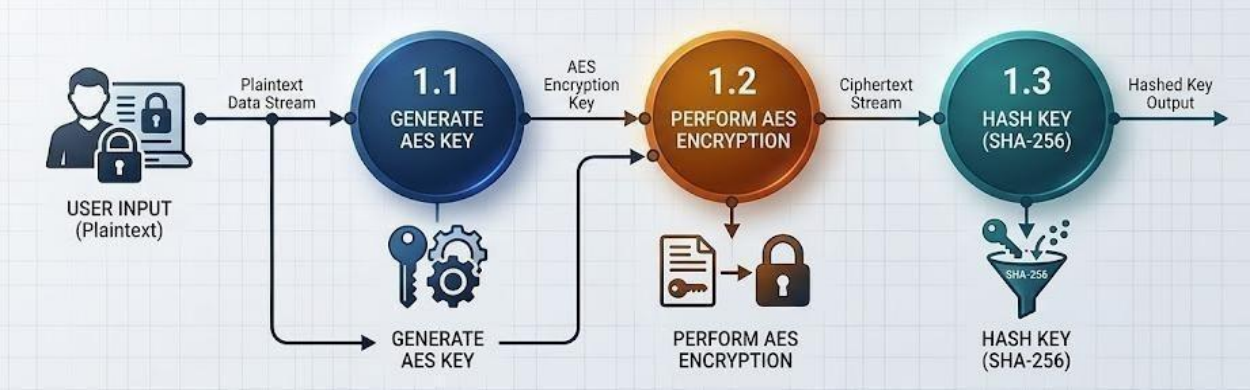


Fig 7.3 - DFD Level 2

The Level 2 DFD provides a microscopic view of the 'Encrypt Payload' process. It details the sequential steps: the input of plaintext, the generation/application of the AES key, the execution of the encryption algorithm, and the parallel generation of the SHA-256 hash for secure key validation prior to transmission. The Level 2 DFD provides a microscopic view of the 'Encrypt Payload' process. It details the sequential steps: the input of plaintext, the generation/application of the AES key, the execution of the encryption algorithm, and the parallel generation of the SHA-256 hash for secure key validation prior to transmission. The Level 2 DFD provides a microscopic view of the 'Encrypt Payload' process. It details the sequential steps: the input of plaintext, the generation/application of the AES key, the execution of the encryption algorithm, and the parallel generation of the SHA-256 hash for secure key validation prior to transmission.

CHAPTER 8

USE CASE DIAGRAM

The Use Case Diagram serves to document the interactions between the external actors and the system's functionalities. In this architecture, the primary actor is the User (Operative). The diagram illustrates the specific, highly secure interaction pathways designed to fulfill the system's objectives. The Use Case Diagram serves to document the interactions between the external actors and the system's functionalities. In this architecture, the primary actor is the User (Operative). The diagram illustrates the specific, highly secure interaction pathways designed to fulfill the system's objectives.



Fig 8.1 – Use Case Diagram

Use Case Diagram depicting the interaction between the User and the System modules.

The above Figure highlights the core operational flow. The User initiates interaction via the 'Login/Register' use case. Once authenticated, they engage with the primary 'Send Message' and

'Upload Media' functions. Crucially, these functions are intrinsically linked to the 'Encrypt/Decrypt' use case, which acts as a mandatory intermediary step. The system autonomously manages the 'Self-destruct message' use case based on TTL parameters, enforcing the ephemeral data lifecycle. Figure 9.1 highlights the core operational flow. The User initiates interaction via the 'Login/Register' use case. Once authenticated, they engage with the primary 'Send Message' and 'Upload Media' functions. Crucially, these functions are intrinsically linked to the 'Encrypt/Decrypt' use case, which acts as a mandatory intermediary step. The system autonomously manages the 'Self-destruct message' use case based on TTL parameters, enforcing the ephemeral data lifecycle. Figure 9.1 highlights the core operational flow. The User initiates interaction via the 'Login/Register' use case. Once authenticated, they engage with the primary 'Send Message' and 'Upload Media' functions. Crucially, these functions are intrinsically linked to the 'Encrypt/Decrypt' use case, which acts as a mandatory intermediary step. The system autonomously manages the 'Self-destruct message' use case based on TTL parameters, enforcing the ephemeral data lifecycle. Figure 8.1 highlights the core operational flow. The User initiates interaction via the 'Login/Register' use case. Once authenticated, they engage with the primary 'Send Message' and 'Upload Media' functions. Crucially, these functions are intrinsically linked to the 'Encrypt/Decrypt' use case, which acts as a mandatory intermediary step. The system autonomously manages the 'Self-destruct message' use case based on TTL parameters, enforcing the ephemeral data lifecycle. The system is designed to provide a secure and efficient interaction environment where each user action is processed through well-defined modules. All operations initiated by the user are validated to ensure correctness and integrity before execution. The encryption mechanism plays a crucial role by safeguarding data prior to transmission, thereby maintaining confidentiality throughout the communication process. Additionally, the system architecture supports smooth interaction flow, enabling users to perform messaging and media-sharing tasks without interruptions. Every message handled by the system is tracked with respect to its lifecycle, ensuring proper enforcement of time-based deletion policies. The integration of automated processes reduces manual intervention and enhances reliability. Furthermore, the system maintains consistency in handling different types of data, including text and multimedia content. Overall, the design ensures a balance between usability and security, providing a robust communication platform.

CHAPTER 9

RESULTS

The developed system successfully demonstrates secure communication through encrypted text and multimedia messaging. All messages are protected using encryption and can only be accessed with the correct cipher key, ensuring data confidentiality. The self-destruct feature effectively deletes messages after the specified TTL, enhancing privacy. Overall, the system provides a reliable, real-time, and secure messaging platform with a smooth user experience.

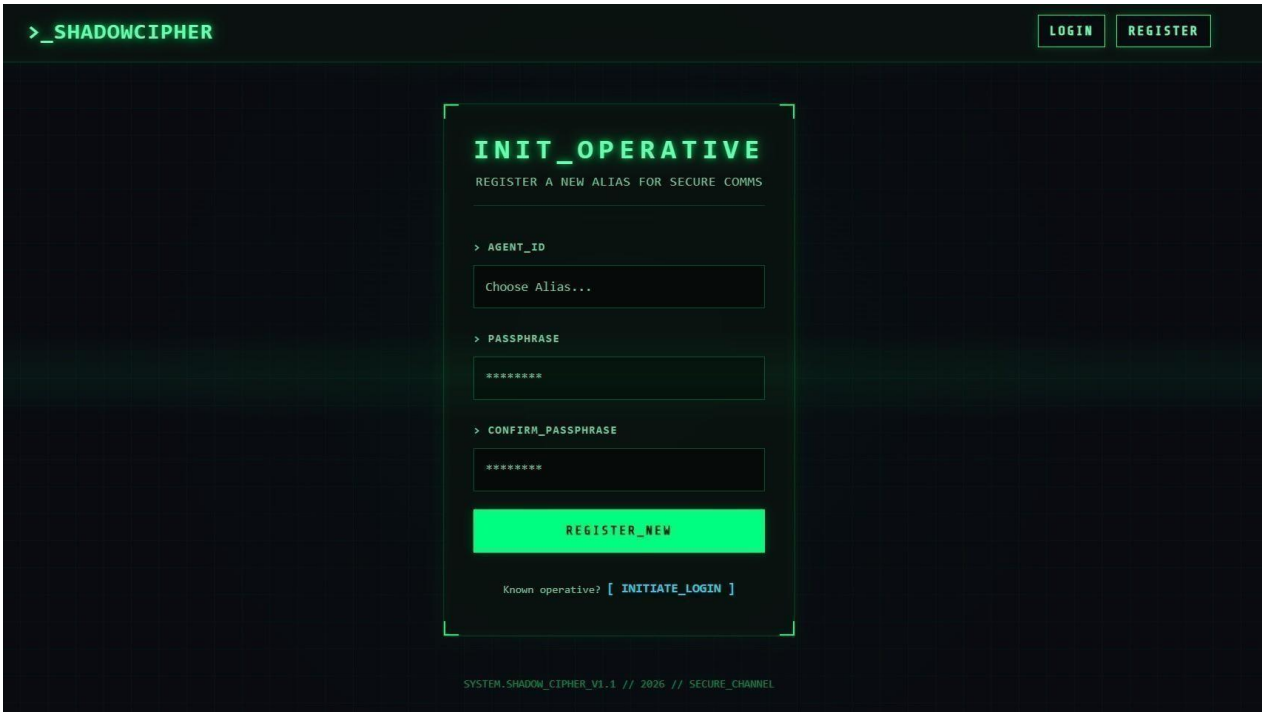


Fig 9.1 – Registration Interface

This screen allows new users to register in the system. It includes fields such as agent ID, passphrase, and confirmation. The registration ensures that each user has unique credentials. This helps maintain system integrity and user authentication. Input validation is implemented to ensure that all required fields are correctly filled before submission. The system verifies that the passphrase and confirmation match to prevent user errors. Upon successful registration, the user is securely stored in the system database. This process ensures a reliable onboarding experience while maintaining strong security standards.

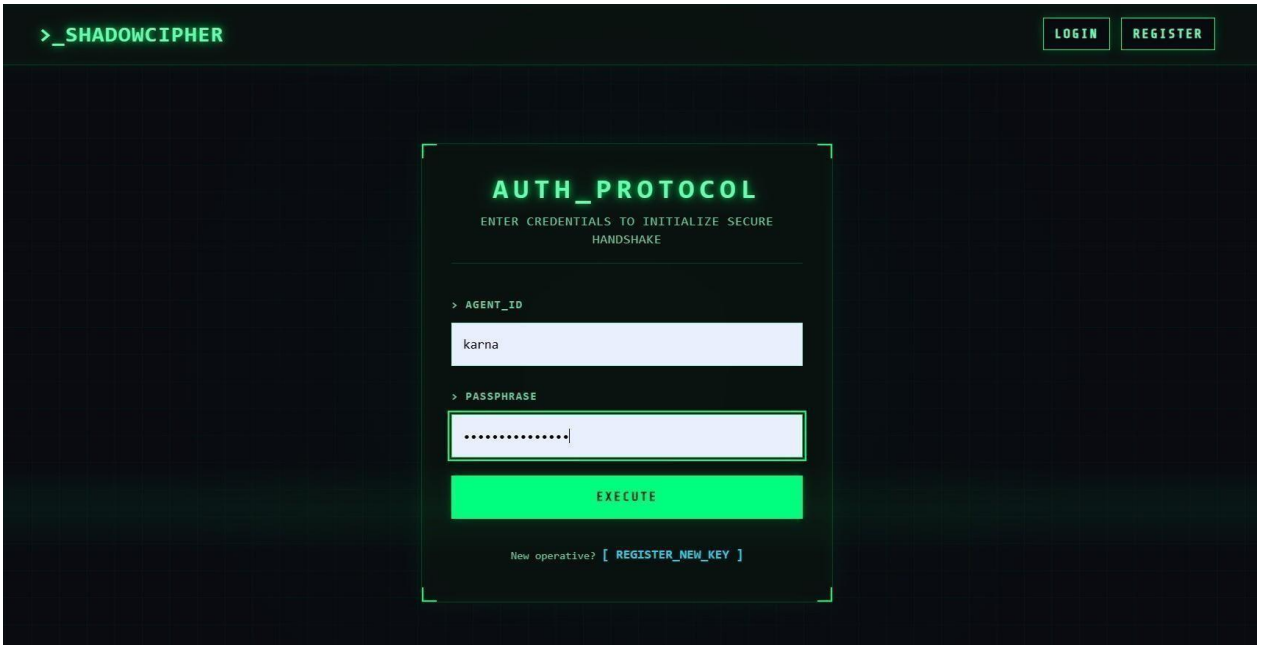


Fig 9.2 – Login (Authentication) Interface

This screen represents the login page where users enter their credentials. It includes fields for agent ID and passphrase. The authentication mechanism ensures secure access to the system. Only valid users can proceed to the messaging dashboard.



Fig 9.3 – Awaiting Connection Screen

This output shows the system waiting for a connection to be established. It prompts the user to select an active link or agent ID. This ensures that communication only happens between authenticated users. It enhances system security and controlled access.

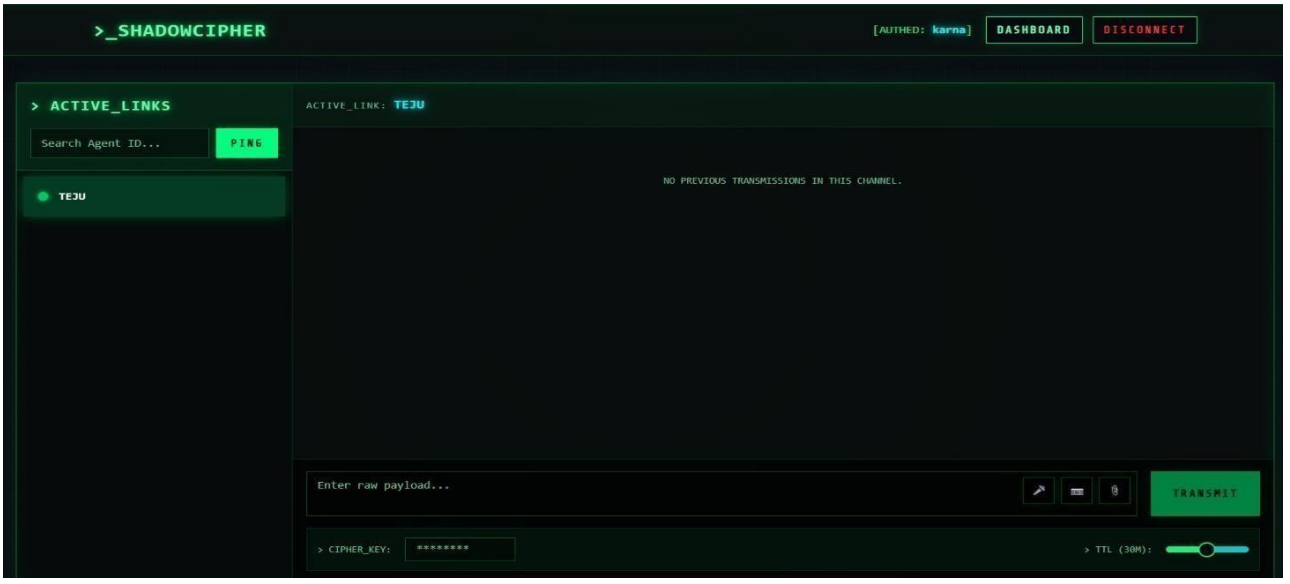


Fig 9.4 – Dashboard – Message Compose Interface

This screen shows the main communication dashboard of the system. The user can type a message in the input field and attach media files using the available options such as microphone, camera, and file upload. The “Transmit” button sends the encrypted payload to the receiver. The interface is designed with a cyber-themed layout for a secure communication feel.

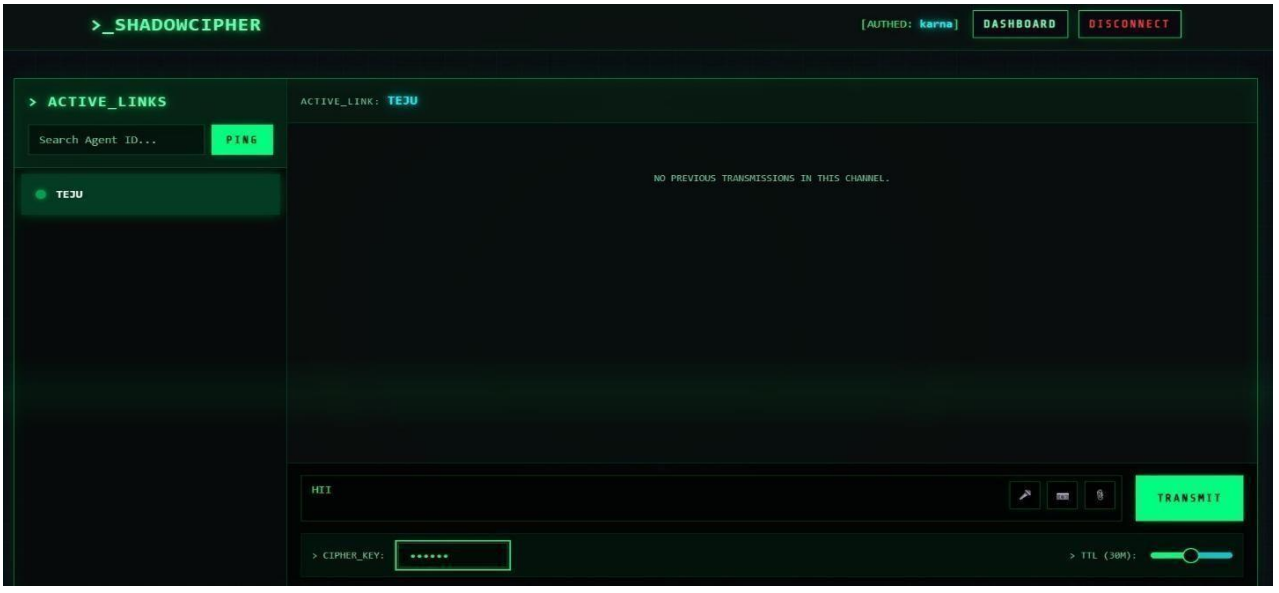


Fig 9.5 – Message Input with Media Options

This screen highlights the input section where users can type messages and attach media. It includes icons for microphone, camera, and file upload. The clean alignment ensures ease of use and quick interaction. This improves overall user experience.

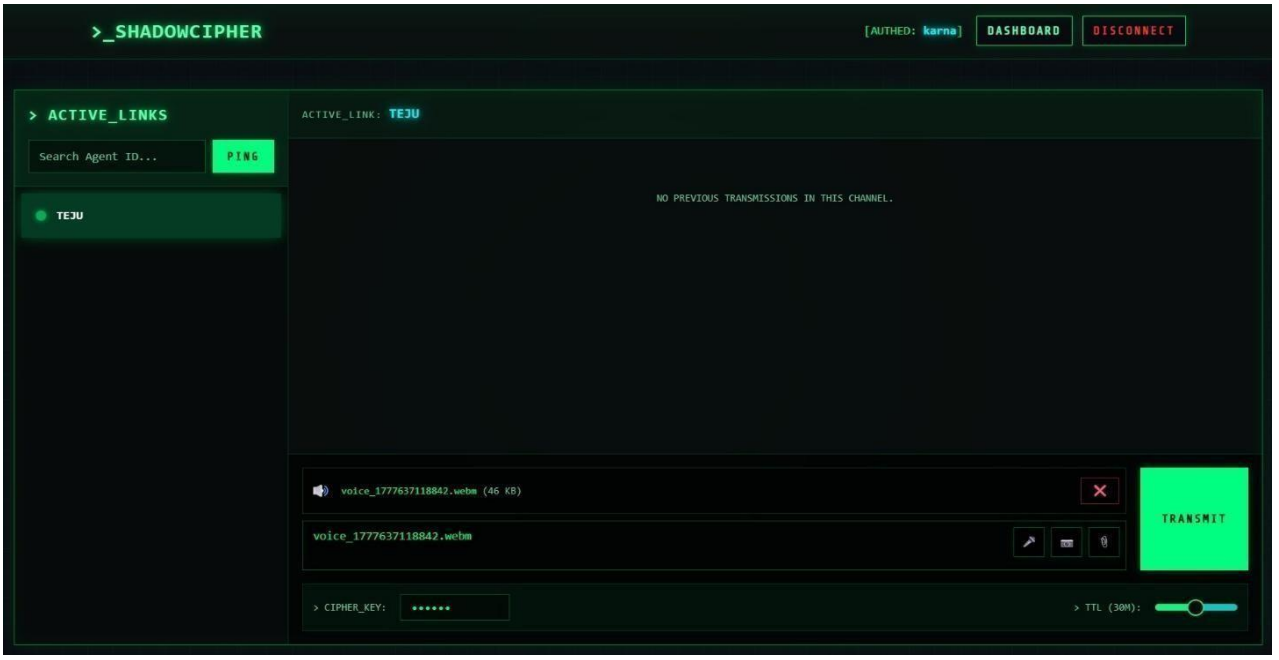


Fig 9.6 – Voice Note Recording

This screen shows the successful recording of a voice message. The recorded audio file is displayed with its file size and name. Users can remove or send the voice note using the provided controls. This enhances multimedia communication within the platform.



Fig 9.7 – Camera Capture Interface

This screen displays the camera feature integrated into the system. The user can capture real-time images using the device camera. The interface includes options such as “Capture” and “Abort.” This allows users to send images securely within the messaging system.

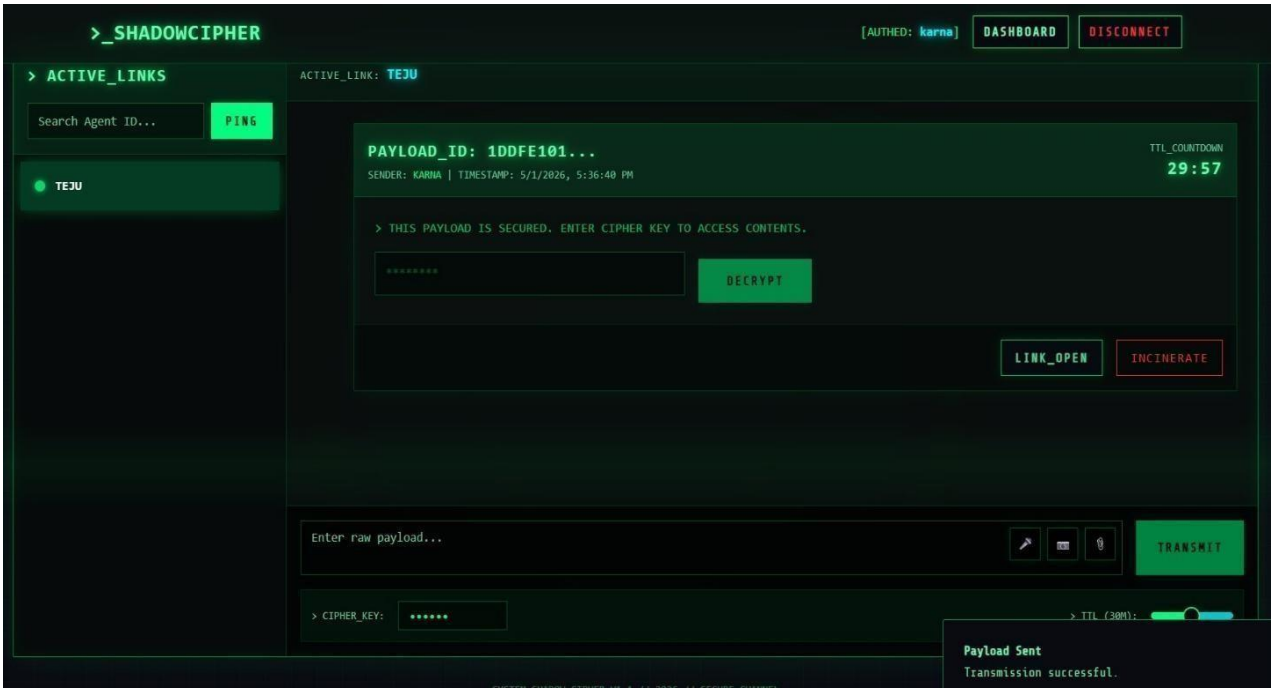


Fig 9.8 – Message Sent Confirmation

This output confirms that a message has been successfully transmitted. A notification such as “Payload Sent – Transmission successful” is displayed to the user. It assures that the encrypted message has been securely delivered to the recipient. This improves user confidence in the system reliability.

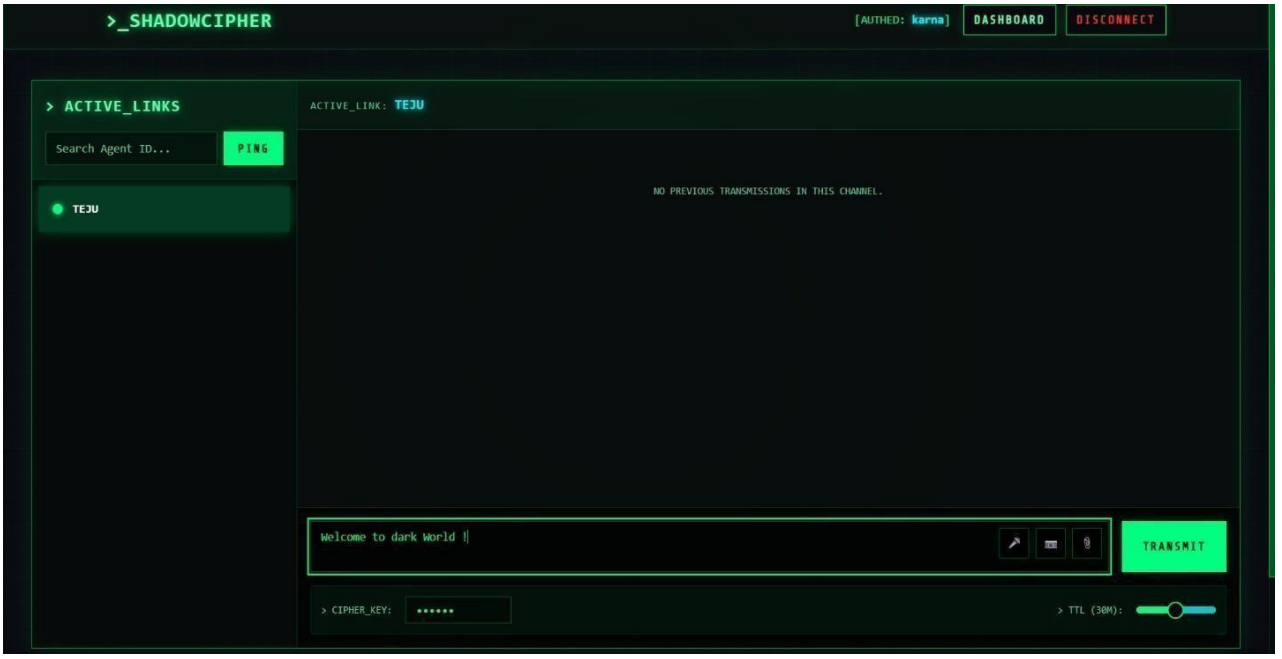


Fig 9.9 – Encrypted Message (Before Decryption)

This screen shows the message in encrypted form before decryption. The user must enter the correct key to access the content. The TTL countdown is visible, indicating message expiry. This ensures data privacy and time-based security.

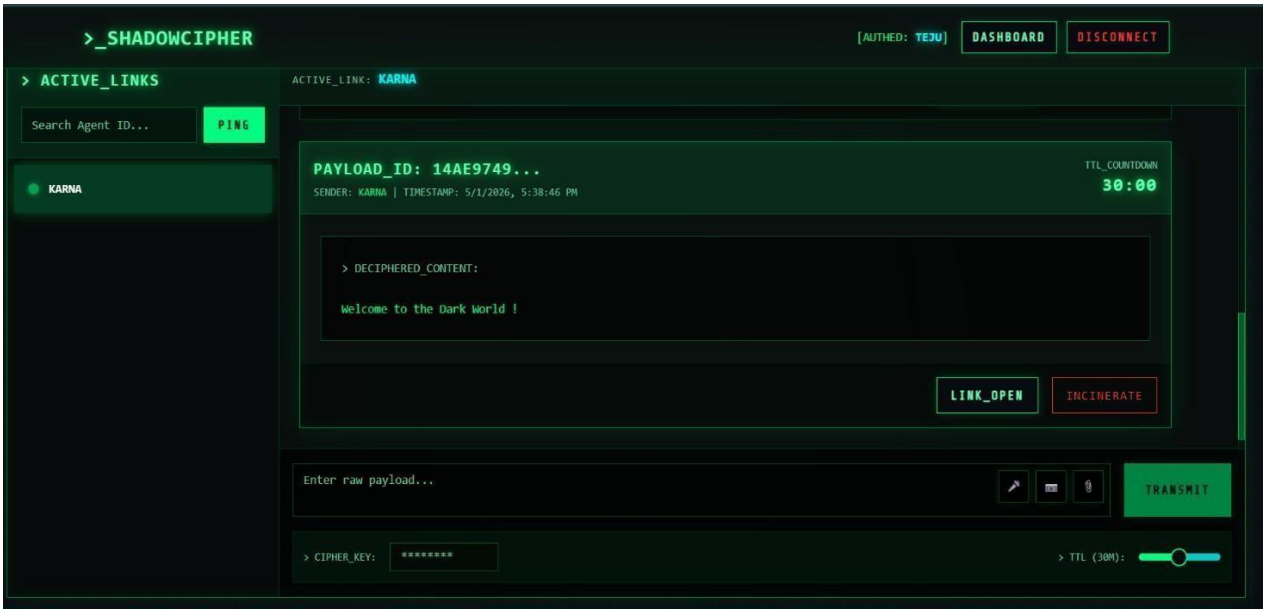


Fig 9.10 – Correct Cipher Key Output Screen

This screen shows the successfully decrypted message after entering the correct cipher key. The system displays the original message content clearly. It also provides options like “Link Open” and “Incinerate.” This confirms secure and correct decryption.

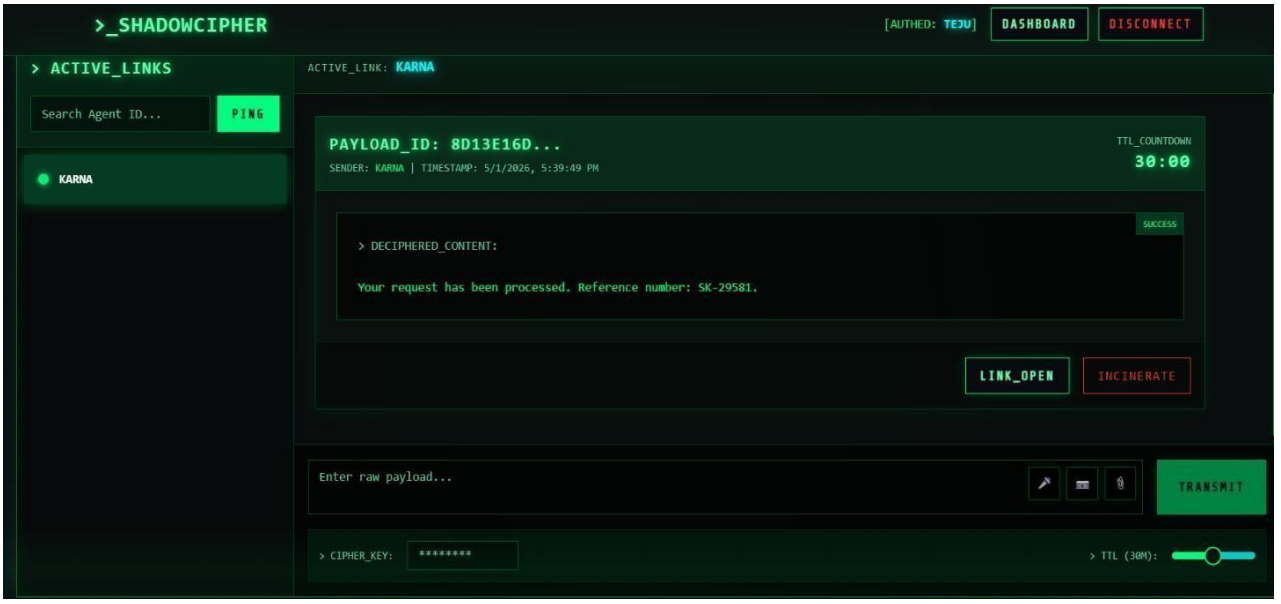


Fig 9.11 – Wrong Cipher Key Output Screen

CHAPTER 10

TESTING AND VALIDATION

Rigorous testing is a fundamental requirement for cryptographic software. A logical failure in an application of this nature does not merely result in a suboptimal user experience; it can lead to catastrophic data exposure. The testing strategy implemented for this project encompasses multiple phases to validate the integrity of the encryption algorithms, the reliability of the real-time transport layer, and the absolute efficacy of the data destruction mechanisms. Rigorous testing is a fundamental requirement for cryptographic software. A logical failure in an application of this nature does not merely result in a suboptimal user experience; it can lead to catastrophic data exposure. The testing strategy implemented for this project encompasses multiple phases to validate the integrity of the encryption algorithms, the reliability of the real-time transport layer, and the absolute efficacy of the data destruction mechanisms. Rigorous testing is a fundamental requirement for cryptographic software. A logical failure in an application of this nature does not merely result in a suboptimal user experience; it can lead to catastrophic data exposure. The testing strategy implemented for this project encompasses multiple phases to validate the integrity of the encryption algorithms, the reliability of the real-time transport layer, and the absolute efficacy of the data destruction mechanisms.

10.1 Testing Methodologies

1. Unit Testing: This phase isolates and verifies individual functions. Specifically, the cryptographic utility functions utilizing `crypto-js` are rigorously tested to ensure that identical plaintext and keys consistently produce predictable ciphertext formats, and that the decryption algorithm perfectly recovers the original string without data loss or corruption.

2. Integration Testing: This phase evaluates the interaction between different system components. Tests confirm that the React Context API correctly manages state, and that the Socket.IO client successfully establishes a connection, emits payloads, and receives broadcasts from the Express backend without connection dropping or data mutation.

3. Functional Testing: This phase simulates end-to-end user workflows. It includes validating the upload of various media types, testing the interaction with device hardware (camera/microphone), and critically verifying the Time-To-Live (TTL) countdown and the subsequent automated deletion of the payload from both the UI and the server's data store.

Test ID	Test Description	Expected Result	Status
TC-01	Attempt authentication with invalid credentials	System rejects access, displays error	PASS
TC-02	Execute AES encryption on text payload	Produces valid Base64 ciphertext string	PASS
TC-03	Attempt decryption with incorrect key	Validation fails (SHA mismatch), decryption aborted	PASS
TC-04	Upload file exceeding size constraints	Upload intercepted, UI displays warning	PASS
TC-05	Monitor TTL expiration sequence	Message irrevocably deleted from Client and Server	PASS
TC-06	Test WebSocket reconnection logic	Client automatically reconnects to server upon drop	PASS
TC-07	Validate camera hardware access	Browser prompts for permission, feed renders correctly	PASS

Fig 10.1.1 – Test Case Table

CHAPTER 11

MODULES IDENTIFIED

The architecture of the system is highly modularized, ensuring separation of concerns, improving code maintainability, and allowing for scalable future development. The application logic is segmented into several primary functional modules. The architecture of the system is highly modularized, ensuring separation of concerns, improving code maintainability, and allowing for scalable future development. The application logic is segmented into several primary functional modules.

Authentication Module:

Manages user registration, secure credential hashing, and session initiation. It acts as the gateway, restricting access to the core application features. Manages user registration, secure credential hashing, and session initiation. It acts as the gateway, restricting access to the core application features. Manages user registration, secure credential hashing, and session initiation. It acts as the gateway, restricting access to the core application features.

Messaging Module:

The central interface component managing the composition, rendering, and interaction with message payloads, including the integration of the TTL countdown UI. The central interface component managing the composition, rendering, and interaction with message payloads, including the integration of the TTL countdown UI. The central interface component managing the composition, rendering, and interaction with message payloads, including the integration of the TTL countdown UI.

Encryption Module

A critical utility module encapsulating the AES encryption, decryption, and SHA-256 hashing logic. It operates entirely on the client-side, enforcing the zero-knowledge paradigm. A critical utility module encapsulating the AES encryption, decryption, and SHA-256 hashing logic. It operates entirely on the client-side, enforcing the zero-knowledge paradigm. A critical utility module encapsulating the AES encryption, decryption, and SHA-256 hashing logic.

Media Module

Interfaces directly with browser hardware APIs to manage live video capture, audio recording, and file system interactions, converting all media to secure Base64 formats. Interfaces directly with browser hardware APIs to manage live video capture, audio recording, and file system interactions, converting all media to secure Base64 formats. Interfaces directly with browser hardware APIs to manage live video capture, audio recording, and file system interactions, converting all media to secure Base64 formats.

Real-time Module

Orchestrates the Socket.IO WebSocket connections, managing the instantaneous, bidirectional transmission of encrypted data and connection state events between clients and the server. Orchestrates the Socket.IO WebSocket connections, managing the instantaneous, bidirectional transmission of encrypted data and connection state events between clients and the server. Orchestrates the Socket.IO WebSocket connections, managing the instantaneous, bidirectional transmission of encrypted data and connection state events between clients and the server.

TTL & Self-Destruct Module:

This module is responsible for managing the time-based lifecycle of messages within the system. It tracks the TTL (Time-To-Live) for each payload and continuously monitors countdown timers. Once the specified time expires, the module automatically deletes the message from both the user interface and the server, ensuring data privacy. This module plays a crucial role in enforcing the ephemeral nature of communication and preventing long-term data storage. The module ensures synchronization between the frontend and backend to maintain consistency in message deletion. It prevents access to expired messages by immediately removing them from active sessions. The countdown timer provides visual feedback to users about the remaining lifetime of each message. This enhances user awareness while strengthening the overall security and privacy of the system.

CHAPTER 12

CONCLUSION

In an era where data breaches, surveillance, and unauthorized access have become significant concerns, the Self-Destructive Encrypted Messenger offers a practical and forward-thinking solution to ensure the confidentiality and impermanence of sensitive digital communications. The system's core strength lies in its use of AES (Advanced Encryption Standard) for data protection and its self-destruction mechanism that ensures messages can be viewed only once before they are automatically deleted, leaving no retrievable trace behind. The integration of secure encryption techniques with a clean, web-based interface enables users to share messages without the risk of long-term data retention or interception. Such a system is particularly relevant in legal, journalistic, corporate, and personal communication environments where data integrity, confidentiality, and user control are critical. Despite its innovation, the system currently operates with certain limitations. Its reliance on a single message-link destruction method, absence of user identity verification, and dependence on server-side operations for message lifecycle management make it vulnerable in highly sensitive scenarios. Furthermore, the lack of mobile app support and limited notification features reduce its real-time usability and responsiveness. Nonetheless, these challenges present clear opportunities for future improvements. Planned enhancements such as the implementation of multi-layered self-destruction triggers, biometric or OTP-based authentication, end-to-end encryption with P2P messaging, and integration with mobile platforms will significantly bolster the system's security and accessibility. Additionally, real-time notifications and activity tracking can improve user awareness and control. In conclusion, this project lays the foundational framework for a robust, privacy-centric communication system. It demonstrates the feasibility and effectiveness of combining encryption and ephemerality to tackle modern data security concerns. With continued development, the Self-Destructive Encrypted Messenger holds the potential to evolve into a comprehensive and industry grade secure messaging platform capable of serving users across various high-stakes and privacy sensitive domains.

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